

SHOBDON AF, United Kingdom

WMO: 035200

Lat: 52.2431N

Lon: 2.8858W

Elev: 325

StdP: 14.52

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.3	(c) 27.2	(d) 21.6	(e) 16.4	(f) 29.6	(g) 24.0	(h) 18.4	(i) 31.1	(j) 29.9	(k) 49.2	(l) 27.2	(m) 49.2	(n) 2.1	(o) 70	(p) 0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.3	(c) 78.3	(d) 66.6	(e) 74.8	(f) 64.7	(g) 71.8	(h) 62.9	(i) 69.2	(j) 73.6	(k) 66.0	(l) 72.1	(m) 63.9	(n) 70.0	(o) 7.1	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 67.9	(b) 103.8	(c) 70.3	(d) 63.8	(e) 89.8	(f) 67.8	(g) 61.6	(h) 82.9	(i) 66.4	(j) 33.4	(k) 73.4	(l) 30.8	(m) 72.2	(n) 29.3	(o) 70.3	(p) 85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	20.2	17.5	17.9	83.3	5.2	3.6	14.1	85.9	11.1	88.0	8.1	90.0	4.3	92.5
			17.9	73.7	5.5	4.2	14.0	76.7	10.8	79.2	7.7	81.5	3.7	84.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	40.0	40.9	43.4	47.2	52.4	57.5	61.0	60.5	56.2	50.9	44.7	40.4
	DBStd	9.17	5.90	5.55	5.16	4.87	5.02	4.49	4.15	4.27	4.20	5.22	5.76	6.93
	HDD50	1452	311	257	211	109	31	2	0	0	3	52	174	302
	HDD65	5632	774	675	670	533	391	227	137	152	265	438	609	761
	CDD50	1325	1	2	7	26	106	228	341	325	190	79	15	5
	CDD65	30	0	0	0	0	0	4	13	12	1	0	0	0
	CDH74	376	0	0	0	0	10	64	186	105	10	1	0	0
	CDH80	48	0	0	0	0	0	7	25	16	0	0	0	0
Wind	WSAvg	6.9	8.0	8.3	8.0	7.0	6.9	6.4	6.0	5.9	5.9	6.5	6.6	7.4
Precipitation	PrecAvg	32.6	3.5	2.6	2.6	2.2	2.4	2.4	1.9	2.6	2.8	3.1	3.2	3.6
	PrecMax	41.2	6.2	5.6	5.8	5.2	6.6	10.1	4.5	7.2	7.8	8.6	7.1	7.4
	PrecMin	22.2	0.4	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.6	0.4	1.1	0.9
	PrecStd	4.3	1.7	1.5	1.2	1.2	1.3	1.9	1.0	1.4	1.7	1.7	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	56.1	61.3	69.5	75.5	80.5	83.1	82.3	75.4	66.3	59.2	56.6
		MCWB	51.8	52.3	52.2	58.1	63.3	66.4	67.0	69.7	67.0	60.1	56.1	53.5
	2%	DB	53.1	53.4	56.9	63.6	70.6	75.0	79.0	76.8	70.4	63.1	57.5	54.6
		MCWB	50.3	50.2	50.7	55.0	61.4	63.8	66.0	67.7	61.8	58.4	54.8	52.1
	5%	DB	51.1	51.6	54.4	59.8	66.3	71.3	75.4	72.7	67.5	61.2	55.7	53.0
		MCWB	48.4	48.4	49.7	52.8	58.3	61.6	65.0	64.2	60.1	57.2	52.9	50.4
	10%	DB	49.4	49.8	52.3	56.6	62.6	67.8	71.7	69.7	65.1	59.4	53.9	50.9
		MCWB	46.7	46.6	48.2	50.6	55.5	59.9	62.9	62.5	58.8	55.8	51.0	48.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	53.1	54.7	59.9	66.9	69.6	74.8	76.7	68.8	62.1	57.0	54.0
		MCDB	54.5	54.9	56.9	66.0	70.1	75.5	75.3	77.2	72.7	65.4	58.6	55.6
	2%	WB	50.7	50.6	52.1	56.2	62.4	65.2	68.5	69.1	63.7	59.7	55.1	52.4
		MCDB	52.7	53.0	55.6	61.7	69.5	72.3	74.8	72.2	68.3	62.1	57.0	54.3
	5%	WB	48.7	48.9	50.4	53.8	59.3	62.8	65.6	65.7	61.7	58.0	53.4	50.8
		MCDB	50.8	51.3	53.7	58.4	64.6	69.3	73.4	70.5	65.8	60.4	55.4	52.7
	10%	WB	47.0	47.0	48.6	51.6	56.9	60.8	63.5	63.6	59.9	56.2	51.4	48.6
		MCDB	49.1	49.5	51.9	55.8	61.4	66.7	70.3	68.5	63.8	58.7	53.5	50.6
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	13.7	16.0	17.0	17.5	18.3	17.7	16.7	13.6	11.7	10.5
		MCDBR	11.4	12.3	17.1	21.8	23.0	24.2	26.1	22.5	21.0	14.8	11.8	10.8
		MCWBR	10.3	9.9	12.3	14.1	14.7	14.3	14.9	14.1	13.7	10.7	10.0	9.8
	5% WB	MCDBR	11.4	12.0	15.6	19.5	20.4	21.0	22.2	19.7	18.4	13.6	11.6	10.8
		MCWBR	10.5	10.4	12.3	14.0	14.8	14.3	15.3	15.5	13.9	10.8	10.3	10.1
Clear-Sky Solar Irradiance	taub		0.318	0.334	0.366	0.389	0.396	0.389	0.386	0.372	0.361	0.345	0.320	0.308
	taud		2.350	2.382	2.293	2.241	2.259	2.306	2.317	2.375	2.407	2.427	2.401	2.322
	Ebn at Noon		214	247	260	267	272	274	272	271	260	239	212	194
	Edh at Noon		20	25	33	39	40	39	38	34	29	24	18	17
All-Sky Solar Radiation	RadAvg		234	443	799	1228	1520	1630	1521	1247	944	532	279	180
	RadStd		21	50	97	142	87	177	162	100	74	49	29	21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-2.34	-2.77	N/A	N/A	N/A	N/A
Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page